BANK FINANCIAL LOAN ANALYSIS SUMMERY

YEARLY loan details.

```sql
SELECT
    COUNT(ID) AS TOTAL_LOAN_APPLICATION,
    SUM(LOAN_AMOUNT) AS TOTAL_FUNDED_AMOUNT,
    SUM(TOTAL_PAYMENT) AS TOTAL_PAYMENT,
    SUM(TOTAL_PAYMENT - LOAN_AMOUNT) AS TOTAL_INTEREST_AMOUNT,
    TO_CHAR(ISSUE_DATE, 'yyyy') AS year
FROM
    BANK_LOAN_DATA
GROUP BY
    year
ORDER BY
    TOTAL_LOAN_APPLICATION DESC;
```

Copy options

	total_loan_application	total_funded_amount bigint	total_payment bigint	total_interest_amount bigint	year text
1	38069	430033125	466809643	36776518	2021

- **Year 2021 ->**
- **TOTAL LOAN APPLICTION – *38,069***
- **TOTAL LOAN AMOUNT – ₹ 43,00,33,125**
- **TOTAL PAYMENT RECEIVED BY BANK - ₹46,68,09,643**
- **TOTAL INTEREST – ₹ 3,67,76,518**
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MONTHLY loan details.

```sql
SELECT
    COUNT(ID) AS TOTAL_LOAN_APPLICATION,
    SUM(LOAN_AMOUNT) AS TOTAL_FUNDED_AMOUNT,
    SUM(TOTAL_PAYMENT) AS TOTAL_PAYMENT,
    SUM(TOTAL_PAYMENT - LOAN_AMOUNT) AS TOTAL_INTEREST_AMOUNT,
    TO_CHAR(ISSUE_DATE, 'MONTH') AS MONTH
FROM
    BANK_LOAN_DATA
GROUP BY
    MONTH
ORDER BY
    TOTAL_LOAN_APPLICATION DESC;
```

	total_loan_application bigint	total_funded_amount bigint	total_payment bigint	total_interest_amount bigint	month text
1	4259	53278500	57277785	3999285	DECEMBER
2	3984	47227075	49592489	2365414	NOVEMBER
3	3741	44257150	48714375	4457225	OCTOBER
4	3497	40533750	43590584	3056834	SEPTEMBER
5	3387	37495000	41910080	4415080	AUGUST
6	3332	35434575	38438852	3004277	JULY
7	3134	33664500	35636982	1972482	JUNE
8	2875	31325000	33322900	1997900	MAY
9	2721	29399100	32063121	2664021	APRIL
10	2588	28343625	31667330	3323705	MARCH
11	2298	24750575	27246915	2496340	JANUARY
12	2253	24324275	27348230	3023955	FEBRUARY

-The bank has received the highest number of loan applications In December month with
- total payment – ₹52277785
Total interest amount - ₹3999285

Quarterly loan details.

```sql
SELECT
    COUNT(ID) AS TOTAL_LOAN_APPLICATION,
    SUM(LOAN_AMOUNT) AS TOTAL_FUNDED_AMOUNT,
    SUM(TOTAL_PAYMENT) AS TOTAL_PAYMENT,
    SUM(TOTAL_PAYMENT - LOAN_AMOUNT) AS TOTAL_INTEREST_AMOUNT,
    DATE_PART('quarter', ISSUE_DATE) AS QUARTER
FROM
    BANK_LOAN_DATA
GROUP BY
    QUARTER
ORDER BY
    TOTAL_LOAN_APPLICATION DESC;
```

	total_loan_application bigint	total_funded_amount bigint	total_payment bigint	total_interest_amount bigint	quarter double precision
1	11984	144762725	155584649	10821924	4
2	10216	113463325	123939516	10476191	3
3	8730	94388600	101023003	6634403	2
4	7139	77418475	86262475	8844000	1

- In quarter 4 bank has received highest loan application with:

- total loan amount – ₹144762725

-total payment – ₹155584649

- total interest amount - ₹10821924

Loan risk status.

```sql
select id ,max(annual_income) as annual_income,
    int_rate ,
            case
                when int_rate > avg(int_rate) over() then 'high risk'
                when int_rate = avg(int_rate) over() then 'medium risk'
                else 'low risk'
            end as  loan_risk_status
from bank_loan_data
group by id , int_rate
order by annual_income desc
limit 10;
```

	id [PK] integer	annual_income integer	int_rate numeric (10,2)	loan_risk_status text
1	513542	6000000	0.13	high risk
2	519954	3900000	0.11	low risk
3	269818	2039784	0.12	low risk
4	611872	1900000	0.06	low risk
5	884755	1782000	0.14	high risk
6	458760	1440000	0.07	low risk
7	453667	1362000	0.08	low risk
8	830027	1250000	0.13	high risk
9	572697	1200000	0.08	low risk
10	603818	1200000	0.15	high risk

Customer income status.

```sql
SELECT id,
       MAX(annual_income) AS annual_income,
       CASE
           WHEN MAX(annual_income) > AVG(annual_income) OVER() THEN 'High Income'
           WHEN MAX(annual_income) = AVG(annual_income) OVER() THEN 'Medium Income'
           ELSE 'Low Income'
       END AS income_status
FROM bank_loan_data
GROUP BY id
ORDER BY annual_income DESC
LIMIT 10;
```

	id [PK] integer	annual_income integer	income_status text
1	513542	6000000	High Income
2	519954	3900000	High Income
3	269818	2039784	High Income
4	611872	1900000	High Income
5	884755	1782000	High Income
6	458760	1440000	High Income
7	453667	1362000	High Income
8	830027	1250000	High Income
9	572697	1200000	High Income
10	603818	1200000	High Income

Total loan application.

```sql
SELECT
    COUNT(*) AS TOTAL_LOAN_APPLICATION
FROM
    BANK_LOAN_DATA;
```

	total_loan_application bigint
1	38069

- **Total 30069 loan Application received by bank.**

Top 5 state where bank has received highest loan application.

	address_state character varying (10)	total_loan_application bigint
1	CA	6780
2	NY	3660
3	FL	2745
4	TX	2620
5	NJ	1804

- **CA on first position from where bank has got highest loan application.**

top 5 emp title where bank have got highest loan application.

```sql
SELECT
    COUNT(ID) AS TOTAL_LOAN_APPLICATION,
    EMP_TITLE
FROM
    BANK_LOAN_DATA
GROUP BY
    EMP_TITLE
ORDER BY
    TOTAL_LOAN_APPLICATION DESC
LIMIT
    5;
```

	total_loan_application bigint	emp_title character varying (50)
1	1424	other
2	126	US Army
3	107	Bank of America
4	66	IBM
5	62	AT&T

- **Where emp title is not describe bank has got highest loan application.**

Highest loan application according to the grade.

```sql
SELECT
    COUNT(ID) AS TOTAL_LOAN_APPLICATION,
    GRADE
FROM
    BANK_LOAN_DATA
GROUP BY
    GRADE
ORDER BY
    TOTAL_LOAN_APPLICATION DESC;
```

	total_loan_application bigint	grade character varying (9)
2	9573	A
3	7788	C
4	5110	D
5	2744	E
6	1005	F
7	309	G

- **Highest loan application in A grade and Grade G has got minimum loan application.**

Sum of loan application and their home ownership status.

```sql
SELECT
    COUNT(ID) AS TOTAL_LOAN_APPLICATION,
    HOME_OWNERSHIP
FROM
    BANK_LOAN_DATA
GROUP BY
    HOME_OWNERSHIP
ORDER BY
    TOTAL_LOAN_APPLICATION DESC;
```

	total_loan_application bigint	home_ownership character varying (50)
1	18198	RENT
2	16970	MORTGAGE
3	2801	OWN
4	97	OTHER
5	3	NONE

- **Where home ownership status is rented bank has got highest loan application.**

Monthly total loan application.

```sql
SELECT
    COUNT(*) AS TOTAL_LOAN_APPLICATION,
    TO_CHAR(ISSUE_DATE, 'month') AS MONTH
FROM
    BANK_LOAN_DATA
GROUP BY
    MONTH
ORDER BY
    TOTAL_LOAN_APPLICATION DESC;
```

	total_loan_application bigint	month text
1	4259	december
2	3984	november
3	3741	october
4	3497	september
5	3387	august
6	3332	july
7	3134	june
8	2875	may
9	2721	april
10	2588	march
11	2298	january
12	2253	february

- **In December bank has got highest loan application and in February month bank has got lowest loan application.**

Total loan application according to the loan status.

```sql
SELECT
    COUNT(ID) AS TOTAL_LOAN_APPLICATION,
    LOAN_STATUS
FROM
    BANK_LOAN_DATA
GROUP BY
    LOAN_STATUS
ORDER BY
    TOTAL_LOAN_APPLICATION DESC;
```

	total_loan_application bigint	loan_status character varying (50)
1	31728	Fully Paid
2	5256	Charged Off
3	1085	Current

- **In this data we have three types of loan status, fully paid loan application is 31728 and current loan application status is 1085.**
- **----------**
- **----------**

Total loan application on loan purpose.

```sql
SELECT
    COUNT(ID) AS TOTAL_LOAN_APPLICATION,
    PURPOSE
FROM
    BANK_LOAN_DATA
GROUP BY
    PURPOSE
ORDER BY
    TOTAL_LOAN_APPLICATION DESC;
```

	total_loan_application bigint	purpose character varying (50)
1	17959	Debt consolidation
2	4927	credit card
3	3776	other
4	2844	home improvement
5	2091	major purchase
6	1757	small business
7	1483	car
8	911	wedding
9	658	medical
10	553	moving
11	361	house
12	346	vacation
13	310	educational
14	93	renewable_energy

- **Where purpose of loan is debt consolidation got 17959 highest loan application.**
- **And renewable energy has got minimum loan application.**

Total loan application for duration.

```sql
SELECT
    COUNT(ID) AS TOTAL_LOAN_APPLICATION,
    TERM
FROM
    BANK_LOAN_DATA
GROUP BY
    TERM;
```

	total_loan_application bigint	term character varying (50)
1	27869	36 months
2	10200	60 months

Total loan application and their verification status.

```sql
SELECT
    COUNT(ID) AS TOTAL_LOAN_APPLICATION,
    VERIFICATION_STATUS
FROM
    BANK_LOAN_DATA
GROUP BY
    VERIFICATION_STATUS
ORDER BY
    TOTAL_LOAN_APPLICATION DESC;
```

	total_loan_application bigint	verification_status character varying (50)
1	16229	Not Verified
2	12181	Verified
3	9659	Source Verified

- **Not verified – highest loan application.**

Total funded amount.

```sql
SELECT
    SUM(LOAN_AMOUNT) AS FUNDED_AMOUNT
FROM
    BANK_LOAN_DATA;
```

	funded_amount bigint
1	430033125

- **Funded amount – which amount bank has funded toward the customers.**

rank of address state according to the funded amount.

```sql
SELECT
    ADDRESS_STATE,
    SUM(LOAN_AMOUNT) AS TOTAL_FUNDED_AMOUNT
FROM
    BANK_LOAN_DATA
GROUP BY
    ADDRESS_STATE
ORDER BY
    TOTAL_FUNDED_AMOUNT DESC
LIMIT
    5;
```

	address_state character varying (10)	total_funded_amount bigint
1	CA	77252500
2	NY	41696175
3	TX	30716450
4	FL	29767500
5	NJ	21409725

-California on first position according to the funded amount.

Monthly Funded amount.

```sql
SELECT
    SUM(LOAN_AMOUNT) AS TOTAL_FUNDED_AMOUNT,
    TO_CHAR(ISSUE_DATE, 'month') AS MONTH
FROM
    BANK_LOAN_DATA
GROUP BY
    MONTH
ORDER BY
    TOTAL_FUNDED_AMOUNT DESC;
```

	total_funded_amount bigint	month text
1	53278500	december
2	47227075	november
3	44257150	october
4	40533750	september
5	37495000	august
6	35434575	july
7	33664500	june
8	31325000	may
9	29399100	april
10	28343625	march
11	24750575	january
12	24324275	february

-total funded amount highest in December (INR-53278500) and lowest in February (INR-24324275).

Total loan application and their funded amount followed by grade.

```sql
SELECT
    GRADE,
    SUM(LOAN_AMOUNT) AS TOTAL_FUNDED_AMOUNT,
    COUNT(ID) AS TOTAL_LOAN_APPLICATION
FROM
    BANK_LOAN_DATA
GROUP BY
    GRADE
ORDER BY
    TOTAL_FUNDED_AMOUNT DESC;
```

	grade character varying (9)	total_funded_amount bigint	total_loan_application bigint
1	B	129228475	11540
2	C	86222825	7788
3	A	83235500	9573
4	D	63047950	5110
5	E	43528600	2744
6	F	18487700	1005
7	G	6282075	309

- **Highest loan amount and loan application in B grade.**
- **Grade – G lowest in this case.**

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OTHER Details: -

```sql
SELECT
    MIN(ISSUE_DATE) AS FIRST_LONE_APPLICATION_DATE,
    MAX(ISSUE_DATE) AS LAST_LOAN_APPLICATION_DATE,
    MIN(LOAN_AMOUNT) AS MIN_FUNDED_AMOUNT,
    MAX(LOAN_AMOUNT) AS MAX_FUNDED_AMOUNT,
    MIN(TOTAL_PAYMENT) AS MIN_TOTAL_PAYMENT,
    MAX(TOTAL_PAYMENT) AS MAX_TOTAL_PAYMENT
FROM
    BANK_LOAN_DATA;
```

	first_lone_application_date date	last_loan_application_date date	min_funded_amount integer	max_funded_amount integer	min_total_payment integer	max_total_payment integer
1	2021-01-01	2021-12-12	500	35000	34	58564

- **Oldest loan application date 2021-01-01**
- **Last loan application on 2021-12-12**
- **Minimum bank's loan amount INR-500.**
- **Maximum loan amount INR- 35000.**
- **Lowest returns INR – ₹ 35.**
- **Maximum returns INR – ₹58564.**

Top highest returns and their loan application ID's.

```sql
SELECT
    ID AS LOAN_ID,
    TOTAL_PAYMENT - LOAN_AMOUNT AS HIGHEST_RETURNS
FROM
    BANK_LOAN_DATA
ORDER BY
    HIGHEST_RETURNS DESC
LIMIT
    5;
```

	loan_id integer	highest_returns integer
1	989285	23564
2	812976	23480
3	972576	22835
4	874599	21849
5	768930	21663

Top 5 loan application and their loan recovery days.

```sql
SELECT
    ID,
    ISSUE_DATE,
    LAST_PAYMENT_DATE,
    (ISSUE_DATE - LAST_PAYMENT_DATE) AS DATE_DIFF,
    MAX(LOAN_AMOUNT) MAX_LOAN_AMOUNT
FROM
    BANK_LOAN_DATA
GROUP BY
    ID,
    DATE_DIFF,
    ISSUE_DATE,
    LAST_PAYMENT_DATE
ORDER BY
    MAX_LOAN_AMOUNT DESC
LIMIT
    5;
```

	id [PK] integer	issue_date date	last_payment_date date	date_diff integer	max_loan_amount integer
1	870019	2021-09-11	2021-04-12	152	35000
2	787951	2021-06-11	2021-09-13	-94	35000
3	679706	2021-02-11	2021-03-16	-33	35000
4	749748	2021-05-11	2021-06-14	-34	35000
5	1047765	2021-12-11	2021-03-12	274	35000

-loan application id '870019' has highest loan recovery period which is 152 days.

Top 5 customer and their total interest amount.

```sql
SELECT
    MAX(INTEREST_AMOUNT) AS HIGHEST_INT_AMOUNT,
    ID AS CUSTOMER ID
FROM
    BANK_LOAN_DATA
GROUP BY
    ID
ORDER BY
    HIGHEST_INT_AMOUNT DESC
LIMIT
    5;
```

	highest_int_amount integer	id [PK] integer
1	23564	989285
2	23480	812976
3	22835	972576
4	21849	874599
5	21663	768930

- **Customer id '989285' has highest interest amount which is INR- 23564 / -**

Highest payment and their recovery periods.

```sql
SELECT
    ID,
    MAX(TOTAL_PAYMENT) AS TOTAL_PAYMENT,
    AGE (ISSUE_DATE, LAST_PAYMENT_DATE) AS RECOVERY_INTERVAL
FROM
    BANK_LOAN_DATA
GROUP BY
    ID,
    RECOVERY_INTERVAL
ORDER BY
    TOTAL_PAYMENT DESC
LIMIT
    5;
```

	id [PK] integer	total_payment integer	recovery_interval interval
1	989285	58564	4 mons 26 days
2	812976	58480	2 mons 26 days
3	972576	57835	7 mons 23 days
4	874599	56849	3 mons 27 days
5	768930	56663	2 mons 26 days

- **On total amount ‘₹ 58564’ and Id ‘989285’ have highest recovery period.**